

Plan 2 / Route δ : Concrete Proof Program for Sharp Quantitative Faber–Krahn Stability

Abstract

This note records the current concrete route from the Plan 2 level-set machinery to the sharp quantitative Faber–Krahn inequality. It is meant to be a compact map of the proof, not a replacement for the individual building blocks. The load-bearing chain is now:

LEVELSETIDENTITY $\rightarrow$ METRICFRAMEWORK $\rightarrow$ FJCENTERBRIDGE $\rightarrow$ TRIMMEDVELOCITYREPAIR
 $\rightarrow$ WEIGHTEDMETRICTRACE $\rightarrow$ BOUNDARYLAYERTRANSFER $\rightarrow$ GLOBALASSEMBLY.

The centroid and space-time notes remain useful diagnostics, but they are not part of the shortest current proof. The former pointwise lower gradient hypothesis Hyp-G is also no longer a standing assumption: on Fusco–Julin good levels, the low-gradient part of the level surface is self-truncated and paid for by the Cauchy–Schwarz defect D_H .

1 Target and normalisation

The final theorem is the sharp quantitative Faber–Krahn inequality: there is $c_{\text{FK}}(n) > 0$ such that every open $\Omega \subset \mathbb{R}^n$, $0 < |\Omega| < \infty$, satisfies

$$\left(\frac{|\Omega|}{|B_1|} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2, \quad (1)$$

where $|B^*| = |\Omega|$ and

$$\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^n} \frac{|\Omega \Delta (x_0 + B^*)|}{|B^*|}.$$

The exponent 2 is sharp, as shown by smooth volume-preserving ellipsoidal perturbations of the ball.

The Plan 2 core proves first the bounded Saint–Venant stability

$$\mathcal{A}(\Omega)^2 \leq C(n, R) \delta_T(\Omega), \quad \delta_T(\Omega) := E(\Omega) - E(B_1), \quad (2)$$

for $|\Omega| = \omega_n$ and $\Omega \subset B_R$, where $E(\Omega) = -T(\Omega)/2$ and $T(\Omega)$ is torsional rigidity. The global Faber–Krahn statement then follows from the BDV bounded reduction and the Kohler–Jobin transfer, exactly as in the Plan 1 final library.

Throughout the bounded part we fix $|\Omega| = \omega_n$ and write $u = u_\Omega$ for the torsion function, $-\Delta u = 1$ in Ω , $u \in H_0^1(\Omega)$. For almost every level,

$$E_\rho := \{u > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n, \quad d\mu(\rho) := (-t'(\rho)) d\rho.$$

The endpoint used in the final transfer is

$$\rho_\delta := 1 - \kappa \sqrt{\delta_T(\Omega)}. \quad (3)$$

For small deficit, ρ_δ lies in a fixed outer collar $[\rho_c, 1]$, for instance with $\rho_c = 3/4$. This collar localisation is enough for the argument.

2 Step 1: exact level-set deficit identity

The first input is the finite-perimeter identity in `LEVELSETIDENTITY`. For $E_t = \{u > t\}$, $m(t) = |E_t|$, define

$$\alpha(t) := \int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}, \quad \beta(t) := \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1},$$

$$D_H(t) := \alpha(t)\beta(t) - P(E_t)^2, \quad D_I(t) := P(E_t)^2 - n^2 \omega_n^{2/n} m(t)^{2-2/n}.$$

Then for every open set of finite measure,

$$\frac{2}{|\Omega|} (E(\Omega) - E(B)) = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt. \quad (4)$$

Both defects are nonnegative. The proof uses only BV coarea on reduced boundaries, Lipschitz truncation for the flux identity $\int_{\partial^* E_t} |\nabla u| = |E_t|$, and the corrected rearranged second derivative $I'' = -1/\alpha$.

On every fixed collar $\rho \in [\rho_c, \rho_\delta]$, (4) gives

$$\int_{\rho_c}^{\rho_\delta} (D_H(t(\rho)) + D_I(t(\rho))) d\mu(\rho) \leq C(n, \rho_c) \delta_T(\Omega). \quad (5)$$

We also use the Talenti comparison in two elementary forms: $-t'(\rho) \leq 1/n$, and a bad-set estimate

$$|\{\rho \in [\rho_c, \rho_\delta] : -t'(\rho) < \rho_c/(2n)\}| \leq C(n, \rho_c) \delta_T(\Omega).$$

The second estimate is the profile-gap input needed to pass from $d\mu$ -weighted kinetic energy to Lebesgue kinetic energy.

3 Step 2: metric first variation

Let $\mathcal{X}$ be the space of finite-measure L^1 classes modulo translations:

$$d_{\mathcal{X}}([A], [C]) = \inf_{a \in \mathbb{R}^n} |A \Delta (C + a)|.$$

Set $F_\rho := \rho^{-1} E_\rho$ so $|F_\rho| = \omega_n$. For a Borel centre field z_ρ , define

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|}, \quad H_{z_\rho, \rho}(x) := \frac{(x - z_\rho) \cdot \nu_\rho(x)}{\rho} \quad (x \in \partial^* E_\rho).$$

The smooth moving-region calculation in `METRICFRAMEWORK` is unconditional, but the finite-perimeter closure is now recorded only in its truthful post-audit form: conditional on a local upper-gradient recovery theorem, one has

$$|\dot{F}_\rho|_{\mathcal{X}} \leq C(n, \rho_c) \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1} \quad (6)$$

for almost every $\rho \in [\rho_c, 1]$. The previous attempt to deduce (6) directly from Reshetnyak and lower semicontinuity of metric derivatives has been removed, because those tools yield the wrong inequality direction for the required upper bound.

4 Step 3: isoperimetric good levels

Fix a small dimensional threshold $\eta_I > 0$ and set

$$G_I := \{\rho \in [\rho_c, \rho_\delta] : D_I(t(\rho)) \leq \eta_I\}, \quad B_I := [\rho_c, \rho_\delta] \setminus G_I.$$

By (5),

$$\mu(B_I) \leq C \delta_T(\Omega). \quad (7)$$

On G_I , the bridge block `FJCENTERBRIDGE` isolates the exact same-centre Fusco–Julin package needed downstream. Conditional on that same-centre theorem and its measurable-selection input, one chooses one Borel centre z_ρ such that

$$d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C(n, R, \rho_c) D_I(t(\rho)). \quad (8)$$

The same-centre package, together with the finite-perimeter oriented radial divergence estimate proved in `FJCENTERBRIDGE`, gives the homothetic trace bound

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_c) D_I(t(\rho)), \quad (9)$$

where

$$\bar{V}_\rho := \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{P(B_\rho)}{P(E_\rho)}.$$

At this stage the same-centre theorem and the measurable-selection input remain explicit external dependencies; they are no longer hidden inside `WEIGHTEDMETRICTRACE`.

5 Step 4: self-truncated velocity repair

This is the repair replacing the former lower-gradient hypothesis. On a good level write $S_\rho := \partial^* E_\rho$, $f := |\nabla u|$, $P := P(E_\rho)$, and

$$\beta := \int_{S_\rho} f d\mathcal{H}^{n-1} = |E_\rho|, \quad \bar{f} := \beta/P.$$

Split

$$S_\rho = G_\rho \cup L_\rho, \quad G_\rho := \{f \geq \bar{f}/2\}, \quad L_\rho := \{f < \bar{f}/2\}.$$

The weighted variance identity for D_H gives

$$\int_{L_\rho} \frac{1}{f} d\mathcal{H}^{n-1} \leq \frac{4}{\beta} D_H(t(\rho)). \quad (10)$$

Thus the low-gradient part is small in the only measure relevant to normal velocity. On G_ρ , $f \geq \bar{f}/2$ gives the reciprocal variance comparison formerly supplied by a pointwise lower bound. The resulting estimate is

$$\left(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, \rho_c) D_H(t(\rho)), \quad \rho \in G_I. \quad (11)$$

This is the key point: tiny components, cusps, or other low-gradient pieces do not need uniform non-degeneracy. They are discarded at the level surface and charged to D_H itself. The discarded swept volume also satisfies

$$\int_{G_I} \int_{L_\rho} V_\rho d\mathcal{H}^{n-1} d\rho \leq C(n, \rho_c) \delta_T(\Omega).$$

6 Step 5: integrated action and kinetic energy

On G_I , choose the same-centre field z_ρ in (6), set $a = 0$, and use the triangle inequality:

$$|V_\rho - H_{z_\rho, \rho}| \leq |V_\rho - \bar{V}_\rho| + |H_{z_\rho, \rho} - \bar{V}_\rho|.$$

Together with (9) and (11), this gives

$$|\dot{F}_\rho|_{\mathcal{X}}^2 \leq C(n, R, \rho_c)(D_H(t(\rho)) + D_I(t(\rho))) \quad (\rho \in G_I). \quad (12)$$

On B_I , the repaired WEIGHTEDMETRICTRACE no longer invokes the unsupported perimeter-free Lipschitz bound (A0); instead it makes its bad-level kinetic requirements explicit as conditional hypotheses. Combining those hypotheses with (5) and (8) yields the integrated action estimate

$$\int_{\rho_c}^{\rho_\delta} \left(d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2 \right) d\mu(\rho) \leq C(n, R, \rho_c) \delta_T(\Omega). \quad (13)$$

Using the Talenti bad-set estimate for $\{-t'(\rho) < \rho_c/(2n)\}$, the same bound implies the Lebesgue kinetic estimate

$$\int_{\rho_c}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C(n, R, \rho_c) \delta_T(\Omega). \quad (14)$$

7 Step 6: weighted metric trace at the endpoint

Let $L^* := (\rho_\delta - \rho_c)/4$ and $J^* := [\rho_\delta - L^*, \rho_\delta]$. From (13) and the measure lower bound $\mu(J^*) \geq c(n, \rho_c)L^* - C\delta_T$, Markov gives a point $\rho_0 \in J^*$ with

$$d_{\mathcal{X}}(F_{\rho_0}, B_1)^2 \leq C \delta_T(\Omega).$$

For any $\rho \in J^*$, the metric absolute-continuity inequality and (14) give

$$d_{\mathcal{X}}(F_\rho, F_{\rho_0})^2 \leq |\rho - \rho_0| \int_{J^*} |\dot{F}_s|_{\mathcal{X}}^2 ds \leq C \delta_T(\Omega).$$

The triangle inequality therefore yields the endpoint trace

$$d_{\mathcal{X}}(F_{\rho_\delta}, B_1)^2 \leq C(n, R, \rho_c) \delta_T(\Omega). \quad (15)$$

This is exactly where the two defects interact: D_I supplies many levels close to balls, while D_H supplies the kinetic control needed to transport that closeness to the endpoint ρ_δ .

8 Step 7: boundary-layer transfer

By definition of $d_{\mathcal{X}}$ and the scaling $F_\rho = \rho^{-1}E_\rho$, (15) implies

$$\mathcal{A}(E_{\rho_\delta})^2 \leq C(n, R, \rho_c) \delta_T(\Omega), \quad (16)$$

where the asymmetry of E_{ρ_δ} is measured at its own volume $\omega_n \rho_\delta^n$. Since $E_{\rho_\delta} \subset \Omega$,

$$|\Omega \setminus E_{\rho_\delta}| = \omega_n(1 - \rho_\delta^n) \leq C(n) \sqrt{\delta_T(\Omega)}$$

by (3). The elementary superlevel transfer lemma gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\rho_\delta}) + \frac{2}{\omega_n} |\Omega \setminus E_{\rho_\delta}|.$$

Squaring and using (16) gives the bounded sharp Saint-Venant inequality

$$\mathcal{A}(\Omega)^2 \leq C(n, R, \rho_c) \delta_T(\Omega) \quad (17)$$

in the small-deficit regime. For large deficit the trivial bound $\mathcal{A}(\Omega) < 2$ is absorbed by increasing C .

9 Step 8: bounded reduction and Kohler–Jobin

Fix $\rho_c = 3/4$ for the Plan 2 core and then apply the bounded result at the BDV dimensional radius $R_*(n)$. The bounded-to-global reduction imported from Plan 1 gives

$$E(\Omega) - E(B_1) \geq c_{SV}^{\text{glob}}(n) \mathcal{A}(\Omega)^2 \quad (|\Omega| = \omega_n).$$

Scaling gives the corresponding global Saint–Venant estimate at arbitrary volume. The Kohler–Jobin transfer then yields (1) with

$$c_{FK}(n) = \min \left\{ \frac{4L_n c_{SV}^{\text{glob}}(n)}{(n+2)T_n}, \frac{L_n(2^{2/(n+2)} - 1)}{4} \right\},$$

where $L_n = \lambda_1(B_1)$ and $T_n = T(B_1)$.

10 What is load-bearing, and what is not

The load-bearing proof is the eight-step chain above. The current role of the surrounding WIP files is as follows.

File	Role	Status in the proof
<code>WIP_LevelSetIdentity</code>	exact $D_H + D_I$ identity	load-bearing
<code>WIP_MetricFramework</code>	metric first variation	conditional on UGR
<code>WIP_FJCenterBridge</code>	same-centre FJ + radial trace	bridge / conditional input
<code>WIP_TrimmedVelocityRepair</code>	removes Hyp-G(lower)	load-bearing
<code>WIP_WeightedMetricTrace</code>	action/kinetic/endpoint trace	load-bearing modulo named inputs
<code>WIP_BoundaryLayerTransfer</code>	endpoint back to Ω	load-bearing
<code>WIP_GlobalAssembly</code>	BDV reduction + KJ transfer	load-bearing
<code>WIP_CentroidBound</code>	centroid diagnostics	auxiliary
<code>WIP_SpaceTimeIdentity</code>	space-time diagnostics	auxiliary

In particular, the following are not used in the current shortest route: BDV/Cicalese–Leonardi selection, graph entry, Schauder interpolation, nearly-spherical expansions, pointwise radial trace (R), pointwise (B^2), and the pointwise lower-gradient Hyp-G.

11 Audit checklist

The remaining useful audit tasks are now sharply focused:

1. Close the upper-gradient recovery theorem needed for (6), or replace it by a direct finite-perimeter deformation estimate.
2. Supply or cite the exact same-centre Fusco–Julin theorem and the measurable-selection input isolated in `WIP_FJCenterBridge.tex`.
3. Check the self-truncated algebra (10)–(11); it is now the active replacement for Hyp-G.
4. Verify the repaired Talenti profile-gap bad-set estimate used to pass from (13) to (14).
5. Verify the explicit bad-level kinetic hypotheses now stated in `WIP_WeightedMetricTrace.tex`.
6. Check the superlevel transfer from E_{ρ_δ} to Ω and the constant bookkeeping in the BDV/Kohler–Jobin imports.

References

- [1] Plan 2 building block LEVELSETIDENTITY, `WIP_LevelSetIdentity.tex`.
- [2] Plan 2 building block METRICFRAMEWORK, `WIP_MetricFramework.tex`.
- [3] Plan 2 bridge block FJCENTERBRIDGE, `WIP_FJCenterBridge.tex`.
- [4] Plan 2 building block TRIMMEDVELOCITYREPAIR, `WIP_TrimmedVelocityRepair.tex`.
- [5] Plan 2 building block WEIGHTEDMETRICTRACE, `WIP_WeightedMetricTrace.tex`.
- [6] Plan 2 building block BOUNDARYLAYERTRANSFER, `WIP_BoundaryLayerTransfer.tex`.
- [7] Plan 2 building block GLOBALASSEMBLY, `WIP_GlobalAssembly.tex`.
- [8] Plan 1 building block BOUNDEDREDUCTION, `Final/BoundedReduction.tex`.
- [9] Plan 1 building block KOHLERJOBINTRANSFER, `Final/KohlerJobinTransfer.tex`.